

CE



control unit



Mindy A924

Instructions and warnings for the fitter

Istruzioni ed avvertenze per l'installatore

Instructions et recommandations pour l'installateur

Anweisungen und Hinweise für den Installateur

Instrucciones y advertencias para el instalador

Instrukcje i uwagi dla instalatora

Aanwijzingen en aanbevelingen voor de installateur

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A924

mindy A924

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Warnings:

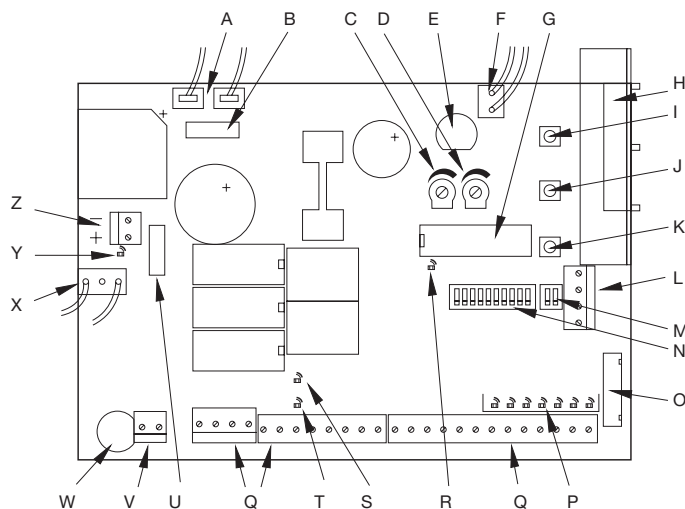
⚠ This manual has been especially written for use by qualified fitters. No information given in this manual can be considered as being of interest to end users! This manual refers to the A924 control unit and may not be used for different products. The A924 control unit has been designed to control the SUMO electromechanical actuator produced by Nice s.p.a.; any other use is considered improper and is consequently forbidden by current law. Do not install the unit before you have read all the instructions.

1) Description of the product

The operating principle of the A924 control unit is based on a position control system featuring a magnetic sensor (encoder) incorporated in the motor. This system detects the degree of rotation of the shaft and offers positioning and speed adjustment functions that traditional control systems cannot achieve. Thanks to the motor speed and torque control, the control unit is able to detect the presence of an obstacle ("current sensitivity function"). The control unit features a manoeuvre counter which allows maintenance operations to be scheduled, is compatible with the radio receivers produced by Nice and is fitted with an internal battery charger.

Fig. 1 offers an overall view of the board and indicates the main components.

- A** Secondary transformer 1 connector
- B** Motor fuse (F2)
- C** Opening manoeuvre current adjustment
- D** Closing manoeuvre current adjustment
- E** Flashing light, electric block, photocell, service fuse (F3)
- F** Secondary transformer 2 connector
- G** Microprocessor
- H** Radio receiver slot
- I** "Open" button
- J** "Memory" button
- K** "Close" button
- L** Aerial terminal and second radio channel
- M** Dip Switch programming
- N** Dip Switch functions
- O** Connector for door controls
- P** Input led
- Q** Motor/input/output terminals
- R** "OK" led
- S** Release led
- T** Encoder led
- U** Battery fuse (F4)
- V** Power input terminals
- W** Line fuse (F1)
- X** Primary transformer connector
- Y** Battery led
- Z** Battery terminal



1

2) Installation

2.1) Preliminary checks

⚠ Automatic gate and door systems may only be installed by qualified fitters in the full respect of the law.

Before starting installation:

- Comply with the warnings shown in the "Warnings for fitters" file.
- Make sure that the mechanical stops can stop the door from moving and that they absorb all the kinetic energy accumulated during movement without difficulty.

2.2) Mounting the A924 control unit

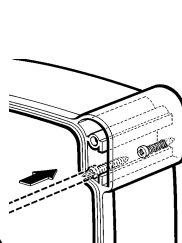
The unit is supplied in a container which, if appropriately installed, will guarantee a protection level of IP55 which means it is also suitable for surface mounted installation.

How to mount the control unit:

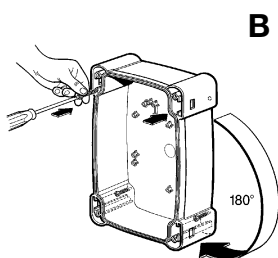
- Install the unit on a permanent surface that is perfectly flat and adequately protected against knocks, making sure the bottom remains at least 40 cm from the ground.
- Install cable or pipe leads only at the bottom of the unit; for no reason whatsoever must the side and top walls be perforated. The

cables must only enter the unit from the bottom!

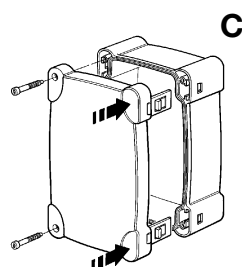
- Fit the two screws into the relative upper holes by sliding them along the guide (**fig. 2A**) and partially tighten them. Rotate the control unit 180° and do the same with the other two screws (**fig. 2B**). Fix the control unit to the wall.
- Fit the cover as required (opening to the left or right) and press hard where shown by the arrows (**fig. 2C**).
- To remove the cover, press on the hinge with a screwdriver and push up (**fig. 2D**).



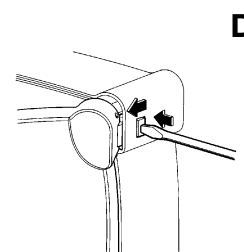
A



B



C



D

2

2.3) Electrical connections

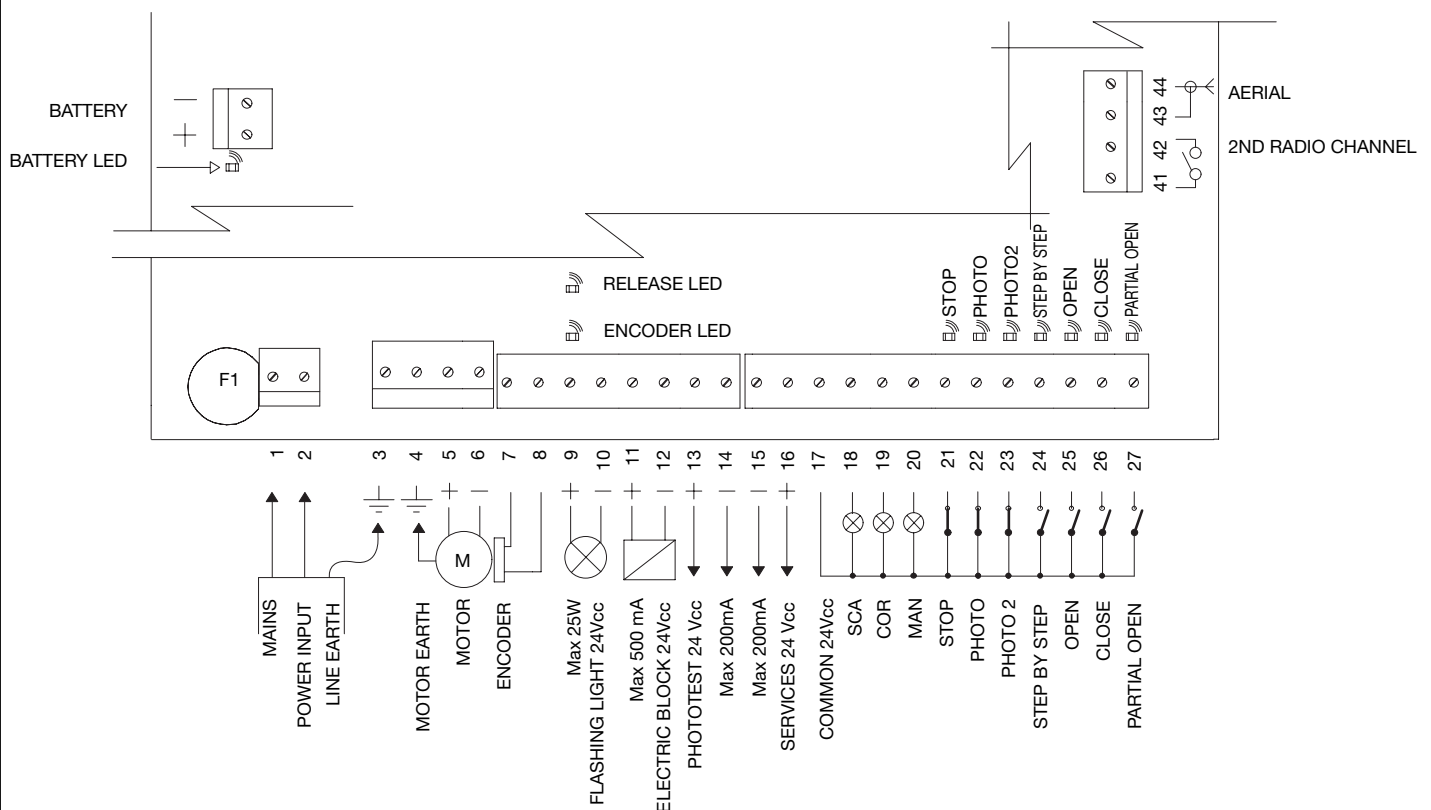
⚠ To protect the operator and avoid damaging the components while electrical connections are being made or the radio receiver is being connected, the control unit may not be mains and/or battery powered

To make connections, please consult the electrical diagram (paragraph 2.4), bearing in mind that:

- The control unit must be powered with a 3 x 1.5mm² cable (phase, neutral and earth); should the distance between the unit and the earth connection exceed 30 m, an earth plate must be fitted near the unit
- For connections to SUMO, a 3x2.5 mm² cable must be used (if over 10m long, use 4mm²) for the motor and a 2 x 0.75mm² cable for the encoder

- Use a cable with a minimum cross-section of 1 mm² to connect the flashing light and the electrical block
- Wires with a minimum cross section of 0.25 mm² must be used to connect very low voltage safety circuits; (use shielded wires if the length exceeds 30 m and connect the earth braid at the control unit end only)
- Maximum attention should be paid to polarised equipment (flashing lamp, electric block, phototest output, services, battery, etc).
- If the inputs of the Normally Closed (NC) contacts are not used they should be jumped with the "24Vdc Common" terminal; if the Normally Open (NO) inputs are not used they should be left free.
- The contacts must be mechanical and potential-free; no stage connections are allowed, such as those defined as "PNP", "NPN", "Open Collector", etc.

2.4) Electrical diagram



2.5) Description of connections

Terminals	Functions	Description
1-2	Phase - Neutral	Mains power input
3	Earth	Control unit earth connection
4	Earth	Motor earth connection
5-6	Motor	36Vdc motor power input
7-8	Encoder	Motor encoder input
9-10	Flashing lamp	24Vdc flashing lamp input max. 25W
11-12	Electrical block/Suction pad	24Vdc electrical block/suction pad output max. 500mA
13-14	Phototest	Phototest output
15-16	24Vdc	24Vdc services power input max. 200mA
17	Common	Common for all inputs
18	Sca	Door open lamp output (on when the door is open, off when it is closed, flashes slowly during the opening manoeuvre, flashes quickly during the closing manoeuvre)
19	Cor	Courtesy lamp output (switches on when the manoeuvre begins and switches off 60 seconds after the manoeuvre has finished)
20	Man	Maintenance lamp output
21	Stop	Input with STOP function (Emergency, shutdown or extreme safety)
22	Photocell	Input for safety devices (photoelectric cells, pneumatic edges) affecting the closing manoeuvre
23	Photo 2	Input for safety devices (photoelectric cells, pneumatic edges) affecting the opening manoeuvre
24	Step-by-step	Input with cyclical Open-Stop-Close-Stop operation
25	Open	Input for opening movement with cyclical Open-Stop-Open-Stop operation
26	Close	Input for closing movement with cyclical Close-Stop-Close-Stop operation
27	Partial open	Input with cyclical Partial Open - Stop - Close- Stop operation
41-42	2° Radio Ch	Output for the second radio receiver channel
43-44	Aerial	Input for the radio receiver aerial
+ -	Battery	24 volt battery connection

2.6) Phototest

The “Phototest” function is an excellent solution as regards the reliability of safety devices and puts the control unit and photocell assembly into category 2 as per UNI EN 954-1 standard (ed. 12/1998).

In order to implement this solution, connect the photocells as shown in one of the figures 3A, 3B or 3C, and move Dip-Switch 7 to On (activate “Phototest”).

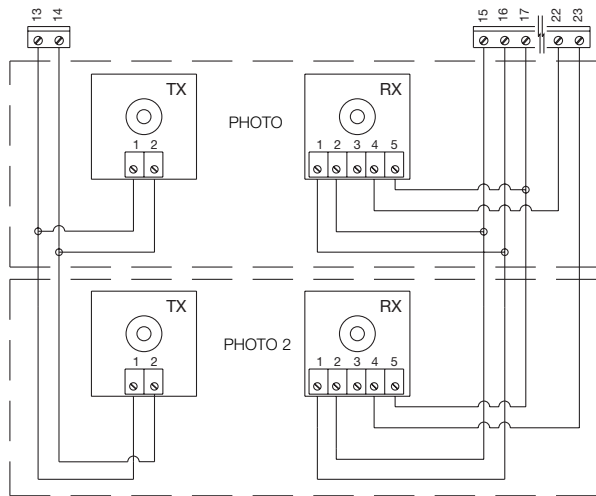


Fig. 3A Shows how to connect Phototest with the Photo and Photo2 photocells

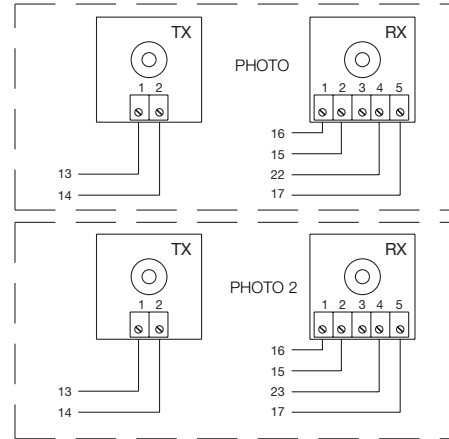


Fig. 3B Shows how to connect Phototest with just the Photo photocell

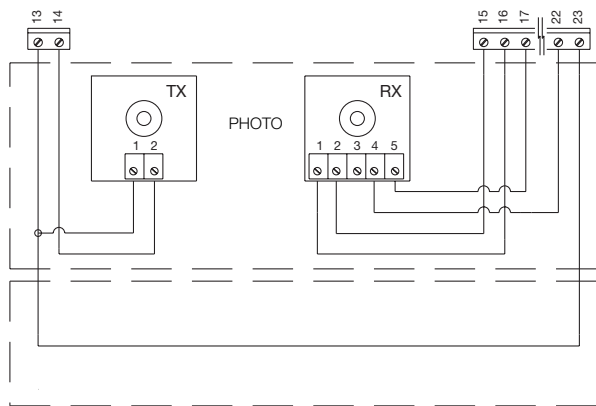


Fig. 3C Shows how to connect Phototest with just the Photo 2 photocell

3

When movement is required, the unit first check that all the receivers involved give their consent, then it turns off the phototest output after which it checks that all the receivers signal the fact by removing their consent; the phototest output is finally reactivated and the

consent of all the receivers is verified once more. If a faulty device or a shorted cable, etc., is detected during the above sequence, the manoeuvre is not carried out.

2.7) Checking connections

⚠ The next operations involve work being done on live circuits, some parts have mains voltage running through them and are therefore extremely dangerous! Pay the greatest of attention to what you are doing and never work alone!

After making connections perform a general test, i.e.:

- Power the control unit and immediately check whether terminals 1-2 are mains powered and terminals 15-16 (services output) are powered at approx. 28Vdc. If this is not the case, unplug the unit immediately and carefully check the connections and the voltage input.
- After about 2 seconds from switching on, the "OK" Led should show the control unit is working correctly by flashing regularly at 1 second intervals.
- Check that the led's relative to the NC inputs ("Stop", "Photo" and "Photo2") are on (safety devices active) and that the led's relative to the NO inputs are off (no command present); if this is not the case, check the connections of the various devices and make sure they are in good working order.
- Check that all the safety devices of the unit are in proper working order (emergency stop, photocells, pneumatic edges, etc.); each time they cut in, the relative "Stop", "Photo" or "Photo2" led's should turn off.
- Check the motor turns in the right direction, i.e.:
 - ♦ release the motor and check that the release led on the board is on
 - ♦ move the door manually so that it is free to move during the opening and closing manoeuvres;
 - ♦ block the motor again and check that the release led is off
 - ♦ press the "Close" button (ref. K of Fig. 1) and check that the door moves in the close direction
 - ♦ if the opening manoeuvre is performed, press again on the close button to stop the door from moving, unplug the machine and invert the two motor wires
 - ♦ regardless of the direction of movement, press the "Close" button again to stop the manoeuvre immediately

3) Programming

Once the connections have been successfully tested, the search for mechanical stops phase can begin.

This operation is important because the A924 control unit must measure the space covered by the motor, in terms of the number of encoder impulses, to move the door from the maximum close position (position 0) to the maximum open position (position 1).

Position 0 and position 1 act as reference points for all the other positions indicated on the graph in **Fig 4**.

POSITION 0: is the point at which the sectional door is in the closed position, corresponding to the mechanical stop (generally the floor).

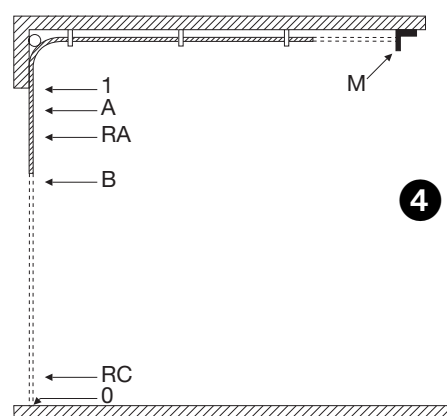
POSITION 1: is the point at which the sectional door is in the maximum open position, corresponding to the mechanical opening stops (point M).

POSITION A: is the point at which the door is required to stop during the opening manoeuvre (this does not necessarily coincide with the mechanical stops in the opening cycle).

POSITION B: is the position in which the door is required to stop during the partial opening manoeuvre.

POSITION RA: is the position in which the door is required to begin slowing down during the normal opening manoeuvre.

POSITION RC: is the position in which the door is required to begin slowing down during the closing manoeuvre.



The mechanical stops can be searched by means of an initial search, an automatic search or manual programming. After an "initial search" or "automatic search", one or more automatically detected positions can be modified by means of manual programming, if required, except positions 0 and 1 which act as references for all the others.

3.1) Initial search for mechanical stops

The "Initial search for mechanical stops" procedure is automatically performed as the first manoeuvre after installation.

Table "A" Activating the initial search for mechanical stops:

1.	Release the motor and move the door manually so that it is free to move during the opening and closing manoeuvres; block the motor.
2.	Briefly press the "Open" button (ref. I, Fig. 1) or the "Close" button (ref. K, Fig. 1) on the board or give a command impulse to the inputs and wait for the control unit to perform a slow closing manoeuvre to position 0, a slow opening manoeuvre to position 1 and a rapid closing manoeuvre to position 0. N.B. If the first manoeuvre after the command is an opening manoeuvre, give another command to stop the procedure and invert motor polarity.
3.	When the above procedure has finished, a mathematical operation is carried out to calculate position A (required opening) at a few centimetres from the maximum open position, position B (partial opening) at about 3/4 from position A, and the RA and RC positions required for slowing down.
4.	The "Initial search" for the mechanical stops has terminated and the gear motor is now ready for use.

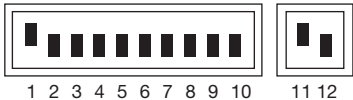
N.B. 1. If one of the safety devices cuts in or another command impulse is given during the "Initial search for mechanical stops", the door will immediately stop moving and the above operations will have to be repeated from the beginning.

3.2) Automatic search for mechanical stops

As an alternative to the "Initial search" the "Automatic search for the mechanical stops" procedure can be performed at any time. This procedure automatically searches for the mechanical stops (position 0 and position 1) in precisely the same way as was described in the initial search section.

Table "B" Activating the automatic search for mechanical stops:

1.	Set the Dip Switches as follows:
----	----------------------------------



1 2 3 4 5 6 7 8 9 10 11 12

2.	Release the motor and move the door manually so that it is free to move during the opening and closing manoeuvres; block the motor.
3.	Press the "Close" button (ref. K, Fig. 1) and wait for the control unit to perform a slow closing manoeuvre to position 0, a slow opening manoeuvre to position 1 and a rapid closing manoeuvre to position 0. N.B. If the first manoeuvre after the command is an opening manoeuvre, give another command to stop the procedure and invert motor polarity.
4.	Using the value of this position, a mathematical operation is carried out to calculate position A (required opening) at a few centimetres from the maximum open position, position B (partial opening) at about 3/4 from position A, and the RA and RC positions required for slowing down.
5.	The "Automatic Search" for the mechanical stops has terminated and the gear motor is now ready for use.

N.B. 1. If one of the safety devices cuts in or another command impulse is given during the "Automatic search for mechanical stops", the door will immediately stop moving and the above operations will have to be repeated from the beginning.

3.3) Memorisation procedure

At the end of the programming sequences of some parameters described in the next few paragraphs, the phase "**Memorisation procedure**" will be seen several times. This operation is used to transfer the value of the parameter required to be programmed into the permanent memory of the control unit.

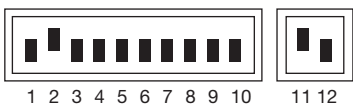
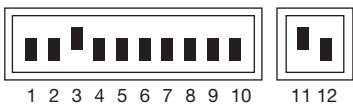
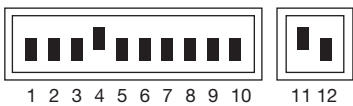
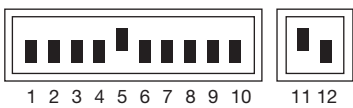
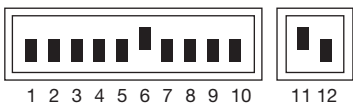
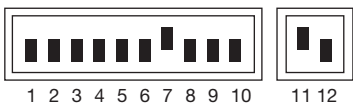
Table "C" Performing the memorisation procedure:

1.	Press the "Memory" button (ref. J fig. 1) for at least 3 s, after 3 s the OK (ref. R, fig.1) led will flash quickly
2.	Release the "Memory" button, the "OK" led will continue to flash quickly for another 3 s.
3.	Within 3 s quickly press the "Open" and "Close" buttons (refs. I and K, fig. 1) together; when these two buttons are pressed together, the ok led will switch off.
4.	Release the two open and close buttons; the "Ok" led will light up for approx. 2 s as confirmation that the memorisation procedure of the selected parameter has terminated correctly

3.4) Manual programming of mechanical stops

This procedure involves manually entering all the positions indicated in **Fig 4**, respecting the order indicated in the following table; in particular, position 0 is the reference position and must be programmed first and never moved.

Table "D" Programming positions manually:

1.	Set the dip switches to one of the following layouts depending on the parameter to memorise
POSITION 0:	Mechanical stop in closing cycle
	
POSITION RC:	Position at which the closing manoeuvre begins slowing down
	
POSITION B:	Stopping position in the Partial opening manoeuvre
	
POSITION RA:	Position at which the normal opening manoeuvre begins slowing down
	
POSITION A:	Stopping position in the Normal opening manoeuvre
	
POSITION 1:	Mechanical stop in opening cycle
	
2.	Press the "Open" button (ref. I, Fig. 1) or "Close" button (ref. K, Fig. 1) on the board and hold it down until the required position is reached.
	N.B. Press the "Memory" button (ref. J, Fig. 1) if acceleration is required.
3.	When the position has been reached, release the buttons and perform the "Memorisation procedure" (paragraph 3.3)
N.B. 1. As an alternative to the manual programming of all the positions, just positions 0, A and 1 may be memorised while positions B, RA and RC can be automatically calculated by the control unit.	

3.5) Programming the electric block position

When the ELB output is assigned an electric block type of function (see switch 8 in paragraph 5) the threshold over which the output switches off can be programmed.

Table "E" Programming electric block positions:

1.	Set the dip switches as shown in the figure:
	
2.	Press the "Open" button (ref. I, Fig. 1) or the "Close" button (ref. K, Fig. 1) until the required position is reached
	N.B. Press the "Memory" button (ref. J, Fig. 1) if acceleration is required.
3.	When the required position has been reached, release the buttons and perform the "Memorisation procedure" (paragraph 3.3)

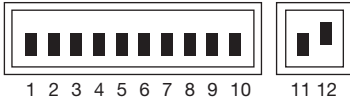
3.6) Programming the pause time

When the automatic close function is selected, a timer is activated after an opening cycle which controls the Pause Time; after this has elapsed, a close manoeuvre is automatically activated. If this time has never been programmed, it is fixed by the control unit at 30 s, but the following procedure allows any value between 1 and 1023 s (approx. 17 minutes) to be programmed.

Table "F" Programming the Pause Time:

1.

Set the dip switches as shown:



2.

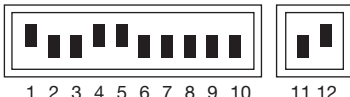
Select the required time with the 10-way dip switch bearing in mind that:

Dip 1 On	activates a pause of	1	s
Dip 2 On	"	2	"
Dip 3 On	"	4	"
Dip 4 On	"	8	"
Dip 5 On	"	16	"
Dip 6 On	"	32	"
Dip 7 On	"	64	"
Dip 8 On	"	128	"
Dip 9 On	"	256	"
Dip 10 On	"	512	"

Example: pause time 25 s

25 = 16+8+1

move dip switches 5 , 4 and 1 to On



If more than one Dip Switch is moved to On, the selected pause time is equal to the sum of the pause times of the single dip switches;

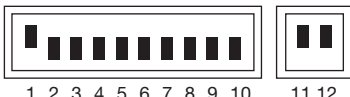
3.

Perform the **"Memorisation procedure"** (paragraph 3.3)

3.7) Deleting the memory

All the programmable parameters are recorded in a permanent memory which stores the information even during a power failure; it may be necessary to delete all the memorised data.

Table "G" Deleting the contents of the memory:

1.	Set the Dip Switches as shown:	
2.	Perform the “memorisation procedure” (see paragraph 3.3) which in this case is used to confirm deletion.	

N.B. When the memory is deleted it is as if the gear motor had never been installed and the door will therefore be unable to move normally; in this case the first command to reach the inputs or pressing the "Open" or "Close" buttons will immediately activate an **"Initial Search for mechanical stops"** procedure.

N.B. 1. This operation does not cancel the number of manoeuvres performed or the number of scheduled manoeuvres.

3.8) Adjustments

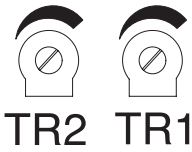
Following the programming phase the few adjustments that are vital for safe and correct operation of the automatic system must be made

3.8.1) Current sensitivity adjustment

To limit the forces brought into play by the movement, a requirement required by the regulations, the control unit features two trimmers TR2 (ref. C, Fig. 1) and TR1 (ref. D, Fig. 1) which allow the cut-in threshold of the current sensitivity device to be varied during the opening manoeuvre and closing manoeuvre respectively. If an obstacle is detected while the door is moving, it stops and, if the “Semiautomatic” or “Automatic” operating mode is active, a manoeuvre in the opposite direction is performed.

If the current sensitivity device cuts in during the closing manoeuvre, the control unit reverses the direction of movement and the opening manoeuvre, unless interrupted, terminates with the door against the mechanical opening stop – position 1 (position reconfirmation)

In order to further increase the safety level, if the current sensitivity device cuts in three times consecutively and prevents the door from closing properly, movement is stopped preceded by a brief inversion.



TR2 = Opening manoeuvre current sensitivity adjustment.
TR1 = Closing manoeuvre current sensitivity adjustment.

3.8.2) Speed adjustment

In order to reduce the kinetic energy released following an impact with an obstacle, as well as adjusting the cut-in threshold of the current sensitivity device the speed of the door during normal operation can also be reduced.

Speed can be adjusted:

- in the “Man present” mode only with the motor stopped,
- in the “Semiautomatic” or “Automatic” mode either with motor stopped or while the door is moving (except for the acceleration and deceleration phases).

Table “H”	Adjusting speed:
1.	Press and hold down the “Memory” button (Ref. J, Fig. 1)
2.	After a second... ♦ press and hold down the “Close” button (ref. K, Fig. 1) to decrease speed, or ♦ press and hold down the “Open” button (ref. I, Fig 1) to increase speed
3.	As soon as the speed has reached the required value, release the buttons (the new speed is automatically memorised)

N.B. The adjustment system is effective until the min. or max. limit values corresponding to the speeds indicated in the motor instructions book are reached; when these limits are reached the “OK” led lights up and remains on when the maximum value has been reached and off when the minimum value has been reached.

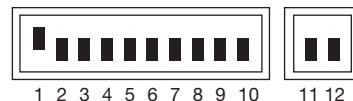
4) Testing

⚠ The automation system must be tested by qualified and expert staff who must establish what tests to perform according to the relative risk.

Testing is the most important part of the whole installation phase. Each single component, e.g. the motor, emergency stop, photocells, etc., may require a specific test phase; please follow the procedures shown in the respective instructions manuals.

To test the A924 control unit, perform the following operations:

1. Set the Dip Switches as shown:
(all the functions deactivated and semiautomatic operation)
2. Press the "Open" button (ref. I, fig. 1) and check that:
 - the flashing lamp activates
 - an opening manoeuvre starts with the acceleration phase
 - the door stops moving, preceded by a deceleration phase, when it reaches position A.
3. Press the "Close" button (ref. K, fig. 1) and check that:
 - the flashing lamp activates
 - a closing manoeuvre starts
 - the door stops moving when it reaches position O.
4. Perform an opening manoeuvre and check that the cut-in of a device connected to the:
 - "Stop" input immediately stops the door moving
 - "Photo" input has no effect
 - "Photo2" input stops and inverts the manoeuvre
5. Perform a closing manoeuvre and check that the cut-in of a device connected to the:
 - "Stop" input immediately stops the door moving
 - "Photo" input stops and inverts the manoeuvre
 - "Photo2" input has no effect
6. Engage a device connected to the:
 - "Stop" input and check that no manoeuvre is performed when a command input is activated
 - "Photo" input and check that no manoeuvre is performed when a command input causing a closing manoeuvre is activated
 - "Photo2" input and check that no manoeuvre is performed when a command input causing an opening manoeuvre is activated
7. During both the opening and the closing manoeuvres, stop the door from moving by introducing an obstacle and check that the manoeuvre inverts before the force indicated by law is exceeded.
8. Check that the activation of the inputs (of connected) causes a step in the sequence
 - for the "Step-by-Step" input: Open – Stop – Close –Stop,
 - for the "Open" input: Open – Stop – Open –Stop,
 - for the "Close" input: Close – Stop – Close –Stop,
 - for the "Partial Open" input: Partial Open – Stop – Close –Stop,



5) Selectable functions

When dip switch programming is Off, the dip switch function allows various other functions to be selected, as shown below:

Switch	1-2	Off Off	=	"Manual" function, i.e.: Man Present
		On Off	=	"Semiautomatic" function
		Off On	=	"Automatic" function, i.e.: Automatic Closing
		On On	=	"Automatic + Always Closes" function
Switch	3	On	=	Condominium function <Not available in the Manual mode>
Switch	4	On	=	5 s pre-flashing (2 s in the manual mode)
Switch	5	On	=	Close again 5 s after Photo in the automatic mode or close again after Photo in the semiautomatic mode
Switch	6	On	=	"Photo" also in the opening manoeuvre
Switch	7	On	=	Activate Phototest
Switch	8	On	=	Suction pad/Electrical block (On = suction pad Off = electrical block)
Switch	9	On	=	Traffic light in the "one way" mode
Switch	10	On	=	Traffic light in the "two way" mode

N.B. Naturally, if a Switch is "Off" its associated function is not activated.

5.1) Description of functions

Man present Function

A movement is only made after a command is given. The Movement stops as soon as the command terminates or after a safety device triggers ("Stop", "Photo" or "Photo2") or the current sensitivity device cuts in. As soon as the movement stops, the command must be stopped before another movement can begin.

Semiautomatic and automatic function

In the "Semiautomatic" or "Automatic" functions, following a command impulse the whole movement is performed until the required position is reached. A second impulse on the input that began the movement will cause it to stop. If, instead of a impulse to a command input a continuous signal is maintained, a state of priority will be created in which the other command inputs are disabled (useful when connecting an opening timer, for example).

If the current sensitivity device or a photocell involved in the direction of movement ("Photo" when closing, "Photo2" when opening) cuts in during a manoeuvre, the direction of movement will be inverted.

In the "Automatic" operating mode, an opening manoeuvre is followed by a pause and then a closing manoeuvre.

If "Photo" triggers during the pause, the timer will be reset with a new pause time; if, on the other hand, the Stop input triggers during the pause, the closing function will be cancelled and the system will Stop.

Always close function

Automatically starts a closing manoeuvre, preceded by 5 s of pre-flashing, if a door is found to be open when power is resumed.

Condominium function

In the "Condominium" function, an opening manoeuvre cannot be interrupted by command impulses except for those triggering a closing manoeuvre. During a closing manoeuvre, a new command pulse will stop the gate and reverse the direction of movement in order to open the gate.

Pre-flashing

Following a command impulse, the flashing lamp is triggered first and then, 5 seconds later (2 seconds later in the manual mode) the manoeuvre begins.

Close again 5 s after Photo in the automatic mode or close again after Photo in the semiautomatic mode

If Photo triggers in the automatic mode during an opening or closing manoeuvre the pause time is reduced to 5 s regardless of the programmed pause time.

If Photo triggers in the semiautomatic mode during a closing manoeuvre the automatic closing manoeuvre is activated with the programmed pause time.

Photo also in the opening manoeuvre

With this function, if the "Photo" safety device triggers, movement will also be interrupted in the opening manoeuvre; if the "Semiautomatic" or "Automatic" functions are selected, the opening movement will continue as soon as "Photo" disengages.

Activate Phototest

This function tests the photoelectric cells before each manoeuvre begins, thereby increasing system safety.

For further details, please consult paragraph 2.6 Phototest

Suction pad/electrical block

This function assigns the following functions to the Elb output (terminals 11 and 12):

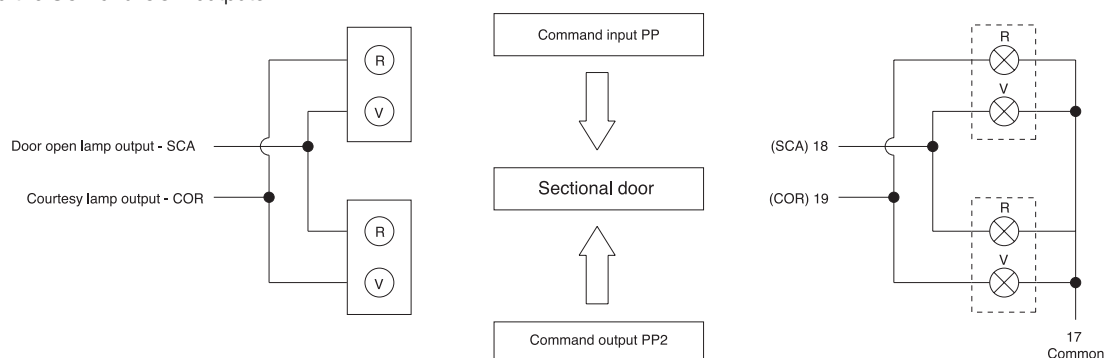
- electrical block (Switch 8 Off) - the output activates during the opening manoeuvre starting from the closed door and remains active until the door exceeds the electrical block position (programmable, see paragraph 3.5);
- suction pad (Switch 8 On) - the output activates at the end of the closing manoeuvre and remains active for the whole time the door is closed.

Traffic light in the one way mode

In this mode, the SCA output is active when the door is open and remains on during the opening manoeuvre while it switches off during the closing manoeuvre and when the door is closed. In this way, a green light can be fitted to the output that, when on, indicates the road is clear.

Traffic light in the two way mode

When switch 10 is On, regardless of the position of switch 9, the "traffic light in the two way" mode is activated; the following changes take place in the control unit: the "Open" input becomes "Step-by-Step2", while the two "Courtesy light" (COR) and "Door open led" (SCA) outputs become Green Light in one direction and Green Light in the other, as shown in **Fig. 5**. A different opening command is given for each direction of movement: "Step-by-Step" (PP) for entering and "Step-by-Step2" (PP2) for exiting; two traffic lights are therefore installed with Red and Green signals connected to the SCA and COR outputs.



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The two SCA and COR outputs are normally off and, as a result, so are the traffic lights; when an open command is given with PP, the opening manoeuvre begins while SCA activates at the same time to switch on the green entry light and the red exit light. If instead the open command is given by PP2, the COR output activates the green exit light and the red entry light. The light will remain on during the whole opening phase and the pause phase, if any; during the closing phase, instead, the green and red lights are activated together to indicate there is no longer transit priority. The two outputs can directly control small 24Vdc lamps for a maximum total of 10W per output. If more powerful lamps are required, it is best to use relays piloted by the control unit outputs that in turn command the traffic lights.

6) Servicing

The control unit is an electronic component and therefore needs no particular maintenance; however, the board and the relative connected devices should be periodically checked (at least every 6 months) by carrying out the whole testing procedure (see Chap. 4).

In order to plan the maintenance programme for the whole system, a manoeuvre counter has been fitted to the control unit which increases its value after every opening manoeuvre. The Maintenance (MAN) led flashes to signal this increase. The value of the manoeuvre counter is constantly compared with an alarm threshold (programmable by the fitter) and the warning threshold (automatically set to the alarm threshold less about 6%). When the number of manoeuvres performed exceeds the warning threshold, the maintenance led only flashes during manoeuvres, while if the alarm threshold is exceeded it continues to flash (with the motor stopped and moving) in order to signal that maintenance must be performed. The alarm threshold can be programmed from a minimum value of 200 to a maximum value of 50800 manoeuvres in multiples of 200.

Tabella “I” Programming the alarm threshold

1.

Set the dip switches as shown

1 2 3 4 5 6 7 8 9 10 11 12

2.

Divide the number of manoeuvres to programme by 100 and then by 2

3.

Consult the table to find the combination of Dip Switches giving the same total value as the above number and move the relative Dip Switches to On.

Dip-switch	Sw1	Sw2	Sw3	Sw4	Sw5	Sw6	Sw7	Sw8
Value	1	2	4	8	16	32	64	128

Example: number of manoeuvres to programme: 30,000
Results after dividing: 150

150 = 128 + 16 + 4 + 2,
Dip Switches 5 , 8, 3 and 2 to On

1 2 3 4 5 6 7 8 9 10 11 12

4.

Perform the “Memorisation procedure” (see paragraph 3.3)

After programming the alarm threshold, visualise it in order to check the operation was performed correctly.

Table “L” Visualising the alarm threshold

1.

Set the dip switches as shown:

1 2 3 4 5 6 7 8 9 10 11 12

2.

Move the Dip Switch 1 to On (2,3,4 and 5 to Off), count the number of times the ok led flashes and write the number on a piece of paper (if it is 10, write 0)

3.

Repeat the operation with Dip Switches 2, 3, 4 and 5

4.

Reconstruct the number of manoeuvres as shown in one of the following two examples

	Dip Switch arrangement	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	1 2 3 4 5	Number of manoeuvres
Example n° 1	Number of times “OK” led flashes	10	1	2	10	4	1.204
Example n° 2	Number of times “OK” led flashes	1	4	10	10	7	14.007

Use the same procedure to visualise the number of manoeuvres performed;

Tabella “M” To visualise the number of manoeuvres performed

1.

Set the dip switches as shown:

1 2 3 4 5 6 7 8 9 10 11 12

2.

Count the number of times the “OK” led flashes with Dip Switches 1, 2, 3, 4 and 5 as shown in examples 1 or 2.

N.B. whenever the warning threshold is programmed the number of manoeuvres performed is automatically cancelled.

6.1) Disposal

This product is made from various kinds of material, some of which can be recycled. Make sure you recycle or dispose of the product in compliance with current laws and bye-laws.

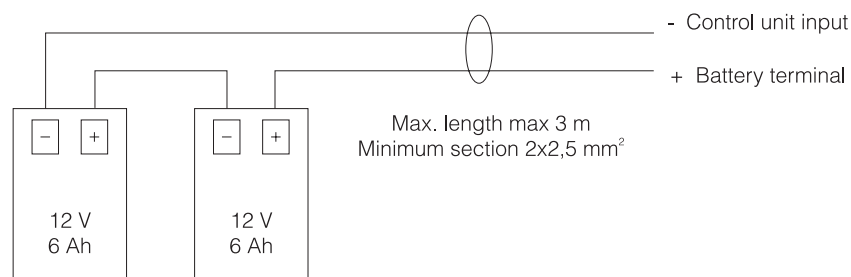
⚠ Some electronic components may contain polluting substances; do not dump them.

7) Battery operation

The control unit contains a large enough transformer to cater for the power requirements of the motor and the electronic board when directly powered by the mains supply. If the system is required to work even when mains power is unavailable, suitable batteries should be connected. When mains power is used, the control unit charges these batteries, while when mains power is unavailable, the control unit automatically switches to the battery mode.

Tabella "N" Connecting the batteries

1. Before connecting the batteries, install the system and make sure everything works correctly.
2. Check that the battery led is on and that voltage at the terminals is approximately 27 volt.
⚠ Perform the next operation with great care as it involves working on live circuits.
3. Connect the batteries as shown in **Fig. 6**.



6

4. Immediately after connecting the batteries to the control unit, check that the battery led is still on; if it is off, disconnect the batteries immediately and check the polarity of the connection.
5. If the battery led is on, check the voltage on the battery terminals again; if voltage is:
 - lower than 18 Vdc, the batteries are not connected correctly or they are faulty
 - between 18 and 25 Vdc, the batteries are flat; wait for them to recharge
 - higher than 25 Vdc, the batteries are charged, disconnect mains power, check the system continues to work correctly and then reconnect mains power.

N.B. In the battery mode, the motor moves more slowly than when the control unit is powered by the mains.

N.B. 1. If both power systems are present (mains power and batteries) and you want to disconnect the power unit for maintenance or safety purposes, as well as disconnecting mains power, you must also disconnect the batteries or remove the F4 battery fuse on the control unit.

8) Radio receiver

The control unit features a connector for plugging in a radio receiver (optional accessory) which activates the Step-by-Step input and allows the control unit to be remote-controlled with a transmitter. The clean contact for the second channel is available on terminal 41-42.

Before fitting the receiver, disconnect mains power and any batteries and plug in the receiver with its components directed towards the control unit microprocessor.

9) What to do if...

Some of the most common problems that can occur during installation are indicated below.

- **No Led is on:**
check terminals 1 and 2 for mains voltage and make sure fuses F1 or F3 have not blown.
- **The manoeuvre does not start:**
check that the safety input led's "Stop", "Photo" and "Photo2" are active and that the motor is blocked (release led off).
- **The direction of movement is inverted during the manoeuvre:**
check that no safety devices have triggered ("Photo" in the closing manoeuvre and "Photo2" in the opening manoeuvre) or that the current sensitivity device has not cut in; in the latter case check that the adjustment is sufficient for the movement of the door. If this is not the case, increase the level by rotating one of the two trimmers clockwise (TR2 for opening, TR1 for closing).
- **The "OK" led flashes quickly:**
the input voltage is insufficient or an incorrect combination has been selected with the Dip switches.
- **The motor moves slowly:**
if it was previously released, the control unit is performing an alignment operation; the first cut-in of the current sensitivity device is considered as a mechanical stop and the correct position is recalled from the memory.
- **The motor performs the acceleration phase and stops:**
check whether the encoder led flashes while the motor is moving. The led may flash at different speeds depending on the speed of movement. When the motor is stopped the led may be on or off depending on the position in which the motor shaft stopped.
- **The "OK" led remains on for a few seconds immediately after a command:**
this shows there is a fault in the motor command stage; check the wiring and earth connection of the motor, other wise replace the control unit.

10) Technical features

Power input	: A924 control unit	➡	230 Vac $\pm$ 10 %, 50-60Hz
	: A924/V1 control unit	➡	120 Vac $\pm$ 10 %, 50-60Hz
Power from batteries	: 21 $\div$ 28 Vdc (two 12 volt batteries, capacity 6Ah)		
Current sensitivity adjustment	: from 2.5 to 15 A		
Service output	: 24 Vdc, maximum current 200mA		
Phototest output	: 24 Vdc maximum current 200 mA		
Flashing lamp output	: 24 Vdc maximum power 25 W		
Electrical block output	: 24 Vdc maximum current 500 mA		
SCA led output	: 24 Vdc maximum power 5W		
Courtesy light output	: 24 Vdc maximum power 5W		
Maintenance led output	: 24 Vdc maximum power 2 W		
Pause Time	: programmable from 1 to 1023 seconds		
Battery charger voltage:	: 27 Vdc		
Charge current	: 200 mA		
Complete charge time	: approx. 24 h for two 12V - 6Ah batteries		
Operating temperature	: -20 a +70 °C		
Protected to	: IP55		
Dimensions and weight	: 220 x 280 h 110, approx. 4 Kg		

9) Wat te doen als....

Dit is bedoeld als een richtlijn voor de installateur om hem te helpen bij het oplossen van de meest voorkomende problemen tijdens de installatie.

- **Er brandt geen enkele ledlampje:**

Controleer of er op de klemmetjes 1 en 2 netspanning aanwezig is en dat de zekeringen F1 of F3 niet gesprongen zijn.

- **De manoeuvre gaat niet van start:**

Verifieer dat de ledlampjes van de ingangen veiligheidsinrichtingen "Stop", "Foto" en "Foto2" actief zijn en dat de motor vergrendeld is (led ontgrendeling uit).

- **Tijdens de manoeuvre vindt er een omkering in de beweging plaats:**

Controleer dat er geen tussenkomst van de veiligheidsinrichtingen ("Foto" bij sluiting en "Foto2" bij opening) of van de stroommeting is geweest; in dit laatste geval dient u na te gaan of de verrichte afstelling voldoende is voor de beweging van de deur. Als dat niet zo is, dient u het niveau te verhogen door één van de twee Trimmers (TR2 voor het openen, TR1 voor het sluiten) met de wijzers van de knop mee te draaien.

- **Het ledlampje "OK" knippert snel:**

De voedingsspanning is onvoldoende of er is met dipschakelaars een verkeerde combinatie geselecteerd.

- **De motor beweegt langzaam:**

Als deze eerder ontgrendeld was, voert de besturingseenheid nu een uitlijning uit; de eerste tussenkomst van de stroommeting wordt beschouwd als een mechanische stop en uit het geheugen wordt de correcte stand opnieuw ingesteld.

- **De motor voert de fase van snelheidstoename uit en stopt:**

Controleer of het ledlampje encoder knippert tijdens de beweging van de motor. De frequentie waarmee het lampje knippert kan groter of kleiner zijn al naar gelang de snelheid van de beweging. Bij stilstaande motor kan het ledlampje aan of uit zijn al naar gelang het punt waar de motoras tot stilstand is gekomen.

- **Ledlampje "OK" brandt onmiddellijk na een instructie enige seconden continu:**

Dit geeft aan dat er een anomalie is in de instructiefase van de motor; controleer de bekabeling en de isolering van de motor; als dit in orde is breng dan een nieuwe besturingseenheid aan.

10) Technische gegevens

Stroomvoorziening	: besturingseenheid A924 ➡	230 Vac ± 10 %, 50-60Hz
	: besturingseenheid A924/V1 ➡	120 Vac ± 10 %, 50-60Hz
Stroomvoorziening via batterij	: 21 ÷ 28 Vcc (2 batterijen van 12 volt, capaciteit 6Ah)	
Afstelling stroommeting	: van 2,5 tot 15 A	
Uitgang neveninrichtingen	: 24 Vcc, max. stroom 200mA	
Uitgang Fototest	: 24 Vcc, max. stroom 200mA	
Uitgang waarschuwingslicht	: 24 Vcc, max. vermogen 25 W	
Uitgang elektrische vergrendeling	: 24 Vcc, max. stroom 500 mA	
Uitgang controlelampje SCA	: 24 Vcc max. vermogen 5W	
Uitgang gebruikerslicht	: 24 Vcc max. vermogen 5W	
Uitgang controlelampje onderhoud	: 24 Vcc max. vermogen 2 W	
Pauseduur	: programmeerbaar van 1 tot 1023 seconden	
Laadspanning batterij	: 27 Vcc	
Laadstroom	: 200 mA	
Duur volledig opladen	: ca. 24 h voor twee batterijen van 12V - 6Ah	
Bedrijfstemperatuur	: - 20 a + 70 °C	
Beschermingsklasse	: IP55	
Afmetingen en gewicht	: 220 X 280 h 110, circa 4 kg	

Dichiarazione CE di conformità / EC declaration of conformity

(secondo Direttiva 98/37/EC, Allegato II, parte B) (according to 98/37/EC Directive, Enclosure II, part B)

Numero / Number: 141/A924

Data / Date: 5/2001

Revisione / Revision: 2

Il sottoscritto Lauro Buoro, Amministratore Delegato, dichiara che il prodotto:

The undersigned Lauro Buoro, General Manager, declares that the product:

Nome produttore / Producer name: NICE s.p.a.
Indirizzo / Address: Via Pezza Alta 13, 31046 Z.I. Rustignè - ODERZO - ITALY
Tipo / Type: Centrale di comando a 1 motore 24Vd.c. per portoni sezionali / Control Unit for 1 motor 24Vd.c. for sectional doors
Modello / Model: A924
Accessori / Accessories: Ricevente radio mod. K, BIO, FLO, FLOR / mod. K, BIO, FLO, FLOR radio receiver

Risulta conforme a quanto previsto dalle seguenti direttive comunitarie / Complies with the following community directives

Riferimento n° Reference n°	Titolo Title
73/23/CEE	DIRETTIVA 73/23/CEE DEL CONSIGLIO del 19 febbraio 1973 concernente il riavvicinamento delle legislazioni degli Stati membri relative al materiale elettrico destinato ad essere adoperato entro taluni limiti di tensione / Council Directive 73/23/EEC of 19 February 1973 on the harmonization of the laws of Member States relating to electrical equipment designed for use within certain voltage limits
89/336/CEE	DIRETTIVA 89/336/CEE DEL CONSIGLIO del 3 maggio 1989, per il riavvicinamento delle legislazioni degli Stati membri relative alla compatibilità elettromagnetica / Council Directive 89/336/EEC of 3 May 1989 on the approximation of the laws of the Member States relating to electromagnetic compatibility
98/37/CE (EX 89/392/CEE)	DIRETTIVA 98/37/CE DEL PARLAMENTO EUROPEO E DEL CONSIGLIO del 22 giugno 1998 concernente il ravvicinamento delle legislazioni degli Stati membri relative alle macchine / DIRECTIVE 98/37/EC OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 22 June 1998 on the approximation of the laws of the Member States relating to machinery

Risulta conforme a quanto previsto dalle seguenti Norme armonizzate / Complies with the following Harmonised standards

Riferimento n° Reference n°	Edizione Issue	Titolo Title	Livello di valutazione Estimate level	Classe Class
EN60335-1	04/1998	Sicurezza degli apparecchi elettrici d'uso domestico e similare - Norme generali. Safety of household and electrical appliances - General requirements		
EN60204-1	09/1993	Sicurezza del macchinario-Equipagg. elettrico delle macchine-Parte 1:Reg.generali Safety of machinery-Electrical equipment of machines-Part 1:General requirements		
EN55022	09/1998	Apparecchi per la tecnologia dell'informazione.Caratteristiche di radiodisturbo. Limiti e metodi di misura/Information technology equipment - Radio disturbance characteristics - Limits and methods of measurement		A
EN55014-1	4/1998	Compatibilità elettromagnetica - Prescrizioni per gli elettrodomestici, gli utensili elettrici e gli apparecchi similari.Parte 1: Emissione- Norma di famiglia di prodotti Electromagnetic Compatibility - Requirements for Household Appliances, Electric Tools and Similar Apparatus - Part 1: Emission - Product Family Standard		
ENV50204	04/1996	Campo elettromagnetico irradiato dai radiotelefonici numerici - Prova di immunità. Radiated Electromagnetic Field from Digital Radio Telephones - Immunity Test	10V/m	A
EN61000-3-2-3	03/1995	Parti 2-3: Armoniche/Flicker/ Parts 2-3: Harmonic/Flicker Compatibilità elettromagnetica (EMC) / Electromagnetic compatibility (EMC) Parte 4: Tecniche di prova e di misura / Part 4: Testing and measurement techniques		A
EN61000-4-2	09/1996	Parte 2: Prove di immunità a scarica elettrostatica / Part 2: Electrostatic discharge immunity test	4KV, 8KV	B
EN61000-4-3	11/1997	Parte 3: Prova d'immunità sui campi irradiati a radiofrequenza Part 3: Radiated, radio-frequency, electromagnetic field immunity test	10V/m	A
EN61000-4-4	09/1996	Parte 4: Test sui transienti veloci/ immunità ai burst / Part 4: Electrical fast transient/burst immunity test	2KV, 1KV	B
EN61000-4-5	06/1997	Parte 5: Prova di immunità ad impulsi / Part 5: Surge immunity test	2KV, 1KV	B
EN61000-4-6	11/1997	Parte 6: Immunità ai disturbi condotti, indotti da campi a radiofrequenza Part 6: Immunity to conducted disturbances, induced by radio-frequency fields.	10V	A
EN61000-4-8	06/1997	Parte 8: Prova di immunità a campi magnetici a frequenza di rete Part 8: Power frequency magnetic field immunity test.	30A/m	A
EN61000-4-11	09/1996	Parte 11: Prove di immunità a buchi di tensione, brevi interruzioni e variazioni di tensione Part 11: Voltage dips, short interruptions and voltage variations immunity tests		B-C

Risulta conforme a quanto previsto dalle altre norme e/o specifiche tecniche di prodotto / Complies with the other standards and/or product technical specifications

Riferimento n° Reference n°	Edizione Issue	Titolo Title	Livello di valutazione Estimate level	Classe Class
EN 12445	11/2000	Industrial, commercial and garage doors and gates / Safety in use of power operated doors - Test methods		
EN 12453	11/2000	Industrial, commercial and garage doors and gates / Safety in use of power operated doors - Requirements		

Inoltre dichiara che non è consentita la messa in servizio del prodotto suindicato finché la macchina, in cui il prodotto stesso è incorporato, non sia identificata e dichiarata conforme alla direttiva 98/37/CE/ The above-mentioned product cannot be used until the machine into which it is incorporated has been identified and declared to comply with the 98/37/CE directive.

Il prodotto suindicato si intende parte integrante di una delle configurazioni di installazione tipiche, come riportato nei nostri cataloghi generali

The above product is an integral part of one of the typical installation configurations as shown in our general catalogues

ODERZO, 16 May 2001

(Amministratore Delegato/General Manager)
Lauro Buoro



COMPANY
WITH QUALITY SYSTEM
CERTIFIED BY DNV
=ISO 9001/2000=

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